Functions: Modulus & Inverse

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The complex number z is defined by $z = \frac{a+3\mathrm{i}}{2+a\mathrm{i}}$, where a is a real constant. Given that $\arg(z) = \frac{\pi}{4}$, find the value of a.

To find a, we first express z in Cartesian form $x + \mathrm{i}y$ by multiplying the numerator and denominator by the complex conjugate of the denominator, $2 - a\mathrm{i}$: $z = \frac{(a+3\mathrm{i})(2-a\mathrm{i})}{(2+a\mathrm{i})(2-a\mathrm{i})} = \frac{2a - a^2\mathrm{i} + 6\mathrm{i} - 3a\mathrm{i}^2}{4+a^2} = \frac{5a + (6-a^2)\mathrm{i}}{4+a^2}$ Thus, the real and imaginary parts are: $\mathrm{Re}(z) = \frac{5a}{4+a^2}$, $\mathrm{Im}(z) = \frac{6-a^2}{4+a^2}$ For.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

1. Find the parameter k: By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k$ $6 + k = 4 \Rightarrow k = -2$ 2. Determine the inverse function $f^{-1}(x)$: With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y+2}{3}$ Thus, $f^{-1}(x) = \frac{x+2}{3}$. 3. Solve the equation $f^{-1}(x) = g(x)$: $\frac{x+2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$ Since the ri.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, find the value of λ.

For a random variable $X \sim \mathrm{Po}(\lambda)$, the probability mass function is given by: $\mathrm{P}(X = x) = \frac{\mathrm{e}^{-\lambda}\lambda^x}{x!}$, $x = 0, 1, 2, \ldots$ Given $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, we substitute $x = 3$ and $x = 4$: $\frac{\mathrm{e}^{-\lambda}\lambda^3}{3!} = \frac{\mathrm{e}^{-\lambda}\lambda^4}{4!}$ Since $\mathrm{e}^{-\lambda} > 0$ and $\lambda > 0$, we can divide both sides by $\mathrm{e}^{-\lambda}\lambda^3$: $\frac{1}{3!} = \frac{\lambda}{4!}$ Multiplyi.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Given that $\cos\theta = -\frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\sin 2\theta$.

We are given $\cos\theta = -\frac{3}{5}$ for an angle in the second quadrant, $\frac{\pi}{2} < \theta < \pi$. 1. In the second quadrant, the sine function is positive: $\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \left(-\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. 2. Use the double-angle identity for sine, $\sin 2\theta = 2\sin\theta\cos\theta$: $\sin 2\theta = 2\left(\frac{4}{5}\right)\left(-\frac{3}{5}\right) = -\frac{24}{25}$. Common misconceptions.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

To solve the modulus equation $|3x - 2| = 2|x + 1|$, we can either consider the linear cases or square both sides. Method 1: Considering linear cases The equation $|A| = |B|$ (where $B = 2(x + 1)$) is equivalent to $A = B$ or $A = -B$: - Case 1: $3x - 2 = 2(x + 1) \Rightarrow 3x - 2 = 2x + 2 \Rightarrow x = 4$ - Case 2: $3x - 2 = -2(x + 1) \Rightarrow 3x - 2 = -2x - 2 \Rightarrow 5x = 0 \Rightarrow x = 0$ Method 2: Squaring both sides Since both sides are non-negative, we can square both sides.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A machine produces steel rods whose lengths are normally distributed with standard deviation $\sigma = 2.4$ cm. A quality controller suspects that the mean length has increased from the target value of 50 cm. A random sample of 36 rods is measured, yielding a sample mean of $\bar{x} = 50.8$ cm. Find the value of the standardized test statistic z for testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.

Under the null hypothesis $H_0 : \mu = 50$, the sample mean of $n = 36$ independent observations follows a normal distribution: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(50, \frac{2.4^2}{36}\right)$ The standard error of the mean is: $\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{2.4}{\sqrt{36}} = \frac{2.4}{6} = 0.4$ The standardized test statistic z is calculated as: $z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{50.8 - 50}{0.4} = \frac{0.8}{0.4} = 2.0$ Common errors include: - O.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The function f is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?

To locate the root using the change of sign method, evaluate $f(x)$ at consecutive integer values: $f(0) = (0)^3 + 3(0) - 5 = -5 < 0$ $f(1) = (1)^3 + 3(1) - 5 = 1 + 3 - 5 = -1 < 0$ $f(2) = (2)^3 + 3(2) - 5 = 8 + 6 - 5 = 9 > 0$ Since $f(x)$ is a continuous polynomial function and there is a sign change between $x = 1$ (where $f(1) < 0$) and $x = 2$ (where $f(2) > 0$), by the Intermediate Value Theorem, a root of $f(x) = 0$ lies between 1 and 2. Checking the other intervals: - On $[0, 1]$, $f(0) = -5$ and.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $P(1 \le X \le 2)$.

To find the probability $P(1 \le X \le 2)$, we integrate the probability density function $f(x)$ over the interval $[1, 2]$: $P(1 \le X \le 2) = \int_1^2 \frac{3}{8}x^2\,dx$ Find the antiderivative: $\int \frac{3}{8}x^2\,dx = \frac{3}{8} \cdot \frac{x^3}{3} = \frac{x^3}{8}$ Evaluate between the limits $x = 1$ and $x = 2$: $\left[\frac{x^3}{8}\right]_1^2 = \frac{2^3}{8} - \frac{1^3}{8} = \frac{8}{8} - \frac{1}{8} = \frac{7}{8}$ Common errors include calculating t.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate y with respect to x using the chain rule. Let $u = 2x^3 + 1$, so $y = \ln(u)$. Then: $\frac{du}{dx} = 6x^2$ and $\frac{dy}{du} = \frac{1}{u}$ Applying the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3 + 1}$ Now evaluate the derivative at $x = 1$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The first term of a geometric progression is 16 and the fourth term is 2. Find the sum to infinity, S_∞, of the progression.

The nth term of a geometric progression is given by $u_n = ar^{n-1}$, where a is the first term and r is the common ratio. Given $a = 16$ and the fourth term $u_4 = 2$: $ar^3 = 2 \Rightarrow 16r^3 = 2 \Rightarrow r^3 = \frac{2}{16} = \frac{1}{8} \Rightarrow r = \frac{1}{2}$ Since $|r| < 1$, the sum to infinity exists and is given by: $S_\infty = \frac{a}{1-r} = \frac{16}{1-\frac{1}{2}} = \frac{16}{\frac{1}{2}} = 32$ Common errors include setting $u_4 = ar^4$ which yields $r^4 = \frac{1}{8}$, or miscalcula.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of all real solutions to the equation $\log_2(x-1) - \log_4(x+14) = 1$

To solve the equation, we first express all logarithms in the same base using the change of base formula: $\log_4(x+14) = \frac{\log_2(x+14)}{\log_2 4} = \frac{\log_2(x+14)}{2} = \frac{1}{2}\log_2(x+14)$ Substitute this back into the original equation: $\log_2(x-1) - \frac{1}{2}\log_2(x+14) = 1$ Multiply through by 2: $2\log_2(x-1) - \log_2(x+14) = 2$ Apply the power law and the quotient law of logarithms.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$

Step 2: Find the parameter k using the given total area of 8.

Area $= \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8: $4\sqrt{3k+4} - 8 = 8 \Rightarrow 4\sqrt{3k+4} = 16 \Rightarrow \sqrt{3k+4} = 4$ Squaring both.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

We solve the problem in three steps: Step 1: Determine q using perpendicularity The direction vectors of l_1 and l_2 are $\mathbf{d}_1 = (1; -2; 2)$ and $\mathbf{d}_2 = (2; 3; q)$. Since $l_1 \perp l_2$, their scalar (dot) product must equal zero: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(2) + (-2)(3) + (2)(q) = 0\ 2 - 6 + 2q = 0 \Rightarrow 2q = 4 \Rightarrow q = 2.$ Step 2: Find the parameters λ and μ at the intersection Equating the coordinates of l_1 and l_2: - x.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

1. Find the parameter λ for a 1-hour period: Let X be the number of calls received in 1 hour. Then $X \sim \text{Po}(\lambda)$. The probability mass function is $\text{P}(X = x) = \frac{\lambda^x \mathrm{e}^{-\lambda}}{x!}$. We are given: $\text{P}(X = 1) = 1.5\,\text{P}(X = 3)$ Substitute the formula for each probability.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Numerical Methods

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\text{P}(X \geq 1 \mid X + Y = 3)$.

Let $X \sim \text{Po}(\lambda)$ and $Y \sim \text{Po}(2\lambda)$ be independent Poisson random variables. The sum $T = X + Y$ follows a Poisson distribution with parameter $\lambda + 2\lambda = 3\lambda$. The conditional distribution of X given $X + Y = n$ is binomial $\text{B}\left(n, \frac{\lambda}{\lambda + 2\lambda}\right) = \text{B}\left(n, \frac{1}{3}\right)$. Here, $n = 3$, so $X \mid (X + Y = 3) \sim \text{B}\left(3, \frac{1}{3}\right)$. We want to find.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Differential Equations

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

A) The standard error of the mean (SE) is calculated using the formula: $SE = \frac{s}{\sqrt{n}}$, where s is the sample standard deviation and n is the sample size. Given $s = 3.5$ grams and $n = 50$. $SE = \frac{3.5}{\sqrt{50}} \approx \frac{3.5}{7.071} \approx 0.495$ grams. B) To construct a 95% confidence interval for the mean weight, we use the formula: $CI = \bar{x} \pm z^* \times SE$, where $\bar{x}$ is the sample mean and z^* is the critical z-value for a 95% confidence level. For a 95% confidence lev.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Poisson Distribution

The curve C has equation $y = \frac{ax + 6}{x^2 + 3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

To find the unknown parameter a, differentiate y with respect to x using the quotient rule: $\frac{dy}{dx} = \frac{a(x^2 + 3) - (ax + 6)(2x)}{(x^2 + 3)^2} = \frac{3a - 12x - ax^2}{(x^2 + 3)^2}$ At a stationary point, $\frac{dy}{dx} = 0$, which requires the numerator to be zero: $3a - 12x - ax^2 = 0$ Given that $x = 1$ is a stationary point: $3a - 12(1) - a(1)^2 = 0 \Rightarrow 2a - 12 = 0 \Rightarrow a = 6$ Substitute $a = 6$ back into the numerator of the derivative.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 13.01 Continuous Random Variables

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

We solve the problem in two steps: Step 1: Find the value of k The Newton-Raphson iteration formula is: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ Differentiating $f(x) = x^3 - 2x^2 - kx + 12$ with respect to x: $f'(x) = 3x^2 - 4x - k$ Evaluating $f(x)$ and $f'(x)$ at $x_0 = 2$: $f(2) = 2^3 - 2(2^2) - 2k + 12 = 8 - 8 - 2k + 12 = 12 - 2k$ $f'(2) = 3(2^2) - 4(2) - k = 12 - 8 - k = 4 - k$ Using the iteration formula for x_1: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow 4 = 2 - \frac{12 - 2k}{4 - k}$ Rearranging.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 14.01 Sampling & Estimation

The complex number z is given by $z = \frac{4+3i}{2-i}$. Express z in the Cartesian form $x + iy$, where x and y are real numbers.

To express z in the form $x + iy$, multiply the numerator and the denominator by the complex conjugate of the denominator. The denominator is $2 - i$, so its complex conjugate is $2 + i$. Multiply numerator and denominator: $z = \frac{(4+3i)(2+i)}{(2-i)(2+i)}$ Expand the numerator: $(4 + 3i)(2 + i) = 8 + 4i + 6i + 3i^2 = 8 + 10i - 3 = 5 + 10i$ Expand the denominator.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 15.01 Hypothesis Testing

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the modulus equation $|3x - 5| = 2x + 1$ for all real solutions.

a. $x = 6$ **and** $x = 0.8$ ✓

b. $x = 6$ only

c. $x = 4$ and $x = -2$

d. $x = 0.8$ only

Case 1: $3x - 5 = 2x + 1 \Rightarrow x = 6$. Check: $|13| = 13$ (valid). Case 2: $3x - 5 = -(2x + 1) \Rightarrow 5x = 4 \Rightarrow x = 4/5 = 0.8$. Check: $|-2.6| = 2.6 = 2(0.8) + 1$ (valid). Both are solutions.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the modulus equation $|2x - 7| = |x + 2|$.

a. $x = 9$ **and** $x = 5/3$ ✓

b. $x = 9$ only

c. $x = 5$ and $x = -3$

d. $x = 3$ and $x = -1$

Square both sides: $(2x-7)^2 = (x+2)^2 \Rightarrow (2x-7-x-2)(2x-7+x+2) = 0 \Rightarrow (x-9)(3x-5) = 0 \Rightarrow x = 9$ or $x = 5/3$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coordinates of the vertex of the graph $y = |2x - 6| + 4$.

a. $(3, 4)$ ✓

b. $(-3, 4)$

c. $(3, -4)$

d. $(6, 4)$

The vertex occurs where the argument of the modulus equals zero: $2x - 6 = 0 \Rightarrow x = 3$. At $x = 3$, $y = 0 + 4 = 4$. Vertex is $(3, 4)$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the inequality $|2x - 5| < 7$.

a. $-1 < x < 6$ ✓

b. $x < -1$ or $x > 6$

c. $-6 < x < 1$

d. $1 < x < 6$

$-7 < 2x - 5 < 7 \Rightarrow -2 < 2x < 12 \Rightarrow -1 < x < 6$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of x satisfying $|x - 2| > 2x - 1$.

a. $x < 1$ ✓

b. $x > 1$

c. $-1 < x < 1$

d. $x < 3$

For $x \geq 2$: $x - 2 > 2x - 1 \Rightarrow x < -1$ (no solution in $[2, \infty)$). For $x < 2$: $-(x - 2) > 2x - 1 \Rightarrow -x + 2 > 2x - 1 \Rightarrow 3x < 3 \Rightarrow x < 1$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $f(x) = 2 + \ln(x - 3)$ is defined for $x > 3$. Find the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = 3 + e^{x-2}$, **domain** $x \in \mathbb{R}$ ✓

b. $f^{-1}(x) = 3 - e^{x-2}$, domain $x > 0$

c. $f^{-1}(x) = 2 + e^{x-3}$, domain $x \in \mathbb{R}$

d. $f^{-1}(x) = \ln(x - 2) + 3$, domain $x > 2$

Let $y = 2 + \ln(x - 3) \Rightarrow \ln(x - 3) = y - 2 \Rightarrow x - 3 = e^{y-2} \Rightarrow x = 3 + e^{y-2}$. The range of f is $\mathbb{R}$, so domain of f^{-1} is $\mathbb{R}$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $f(x) = e^{2x}$ for $x \in \mathbb{R}$ and $g(x) = \frac{1}{x}$ for $x > 0$, solve the equation $fg(x) = e^4$.

a. $x = 1/2$ ✓

b. $x = 2$

c. $x = 1/4$

d. $x = 4$

$fg(x) = f(1/x) = e^{2/x} = e^4 \Rightarrow 2/x = 4 \Rightarrow x = 2/4 = 1/2$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function $h(x) = x^2 - 4x + 1$ is defined on the domain $x \leq k$. Find the greatest value of k for which h is one-to-one.

a. $k = 2$ ✓

b. $k = 4$

c. $k = 0$

d. $k = 1$

The parabola has vertex at $x = -(-4)/(2) = 2$. For $x \leq k$ to be strictly decreasing (one-to-one), the domain must not extend past the vertex, so greatest $k = 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Describe the transformation mapping the graph of $y = \ln(x)$ onto the graph of $y = \ln(3x - 6)$.

a. **Horizontal stretch scale factor** $1/3$ **followed by translation** $[2, 0]$ ✓

b. Translation $[6, 0]$ followed by horizontal stretch scale factor 3

c. Horizontal stretch scale factor 3 followed by translation $[-2, 0]$

d. Vertical stretch scale factor 3 followed by translation $[2, 0]$

Write as $y = \ln[3(x - 2)]$. The factor 3 represents a horizontal stretch by scale factor $1/3$, and $(x - 2)$ represents a translation of 2 units in the positive x-direction ($[2, 0]$).

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the number of real roots of the equation $|2x - 3| = x^2 - 2$.

a. **2 roots** ✓

b. 4 roots

c. 0 roots

d. 1 root

For $x \geq 1.5$: $2x - 3 = x^2 - 2 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$ (not in $x \geq 1.5$). For $x < 1.5$: $-2x + 3 = x^2 - 2 \Rightarrow x^2 + 2x - 5 = 0 \Rightarrow x = \frac{-2 \pm \sqrt{24}}{2} = -1 \pm \sqrt{6}$. $-1 + \sqrt{6} \approx 1.45 < 1.5$ (valid), and $-1 - \sqrt{6} \approx -3.45 < 1.5$ (valid). Exactly 2 roots.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x)$ passes through $(2, 5)$. What are the coordinates of the corresponding point on $y = 3 - 2f(x + 1)$?

a. $(1, -7)$ ✓

b. $(3, -7)$

c. $(1, 13)$

d. $(3, 7)$

New $x = 2 - 1 = 1$. New $y = 3 - 2(5) = 3 - 10 = -7$. Point is $(1, -7)$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ **or** $5 < k < 11$ ✓

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

1. **Find $f^{-1}(x)$ and its domain:** Completing the square gives $f(x) = (x - 3)^2 + 4$. For the domain $x \geq 3$, $f(x)$ is strictly increasing with range $[4, \infty)$. Setting $y = (x - 3)^2 + 4$ with $x \geq 3$ yields $x - 3 = \sqrt{y - 4}$, so $f^{-1}(x) = 3 + \sqrt{x - 4}, \quad \text{with domain } x \geq 4$. 2. **Set up the equation $f^{-1}(x) = |2x - k|$:** Let $u = \sqrt{x - 4} \geq 0$, which implies $x = u^2 + 4 \geq 4$. The equation becomes $3 + u = |2(u^2 + 4) - k| = |2u^2 + 8 - k|$. 3. **Analyz.

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33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$ ✓

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

To find the inverse function $f^{-1}(x)$, set $y = f(x)$ and rearrange to express x in terms of y: $y = \frac{2x+1}{x-3}$ Multiply both sides by $x - 3$: $y(x-3) = 2x + 1$ $yx - 3y = 2x + 1$ Collect all terms containing x on one side and constant/independent terms on the other: $yx - 2x = 3y + 1$ Factor out x: $x(y-2) = 3y + 1$ Divide by $y - 2$: $x = \frac{3y+1}{y-2}$ Interchanging the variables gives: $f^{-1}(x) = \frac{3x+1}{x-2}$ Common Misconceptions: - Choice.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The function $f(x) = (x + 11)^2 - 4$ has domain $x \geq -11$. State the range of f.

a. $f(x) \leq -4$

b. $f(x) \geq -4$ ✓

c. $f(x) \geq -11$

d. $x \geq -11$

Since $x \geq -11$, the minimum of f occurs at $x = -11$, giving $f(-11) = -4$, and f increases thereafter. Range: $f(x) \geq -4$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve $|x - 6| > 6$. What is the smaller boundary value separating the two regions?

a. 11

b. **0** ✓

c. 12

d. 1

$x - 6 > 6$ or $x - 6 < -6$, giving boundary values $x = 12$ and $x = 0$. Smaller boundary: $x = 0$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $|2x + 3| = 3$ for x. What is the larger of the two solutions?

a. **0** ✓

b. -3

c. 3

d. $\frac{1}{12}$

$2x + 3 = \pm 3$, giving $x = 0$ or $x = -3$. The larger solution is $x = 0$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $|2x - 4| < 2$. What is the upper bound of the solution interval?

a. **3** ✓

b. 1

c. 0

d. 4

$-2 < 2x - 4 < 2 \Rightarrow 1 < x < 3$. Upper bound: $x = 3$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Continuous Random Variables

A root of $f(x)$ lies in $[-6, -3]$, where $f(-6)$ is negative and $f(-3)$ is positive. After one bisection step (midpoint $m = \frac{-6-3}{2} = -\frac{9}{2}$), $f(m)$ is negative. Which sub-interval contains the root?

a. $[-6, -\frac{9}{2}]$

b. $[-\frac{9}{2}, -3]$ ✓

c. [-6,-3]

d. [-7,-6]

Since $f(m)$ is negative, the sign change lies in $[-\frac{9}{2}, -3]$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Sampling & Estimation

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$ ✓

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

To find the series expansion of $f(x)$, we first decompose $f(x)$ into partial fractions. **Step 1: Algebraic Division** The degree of the numerator is 4 and the degree of the denominator is $1 + 2 = 3$. Because the numerator has a higher degree than the denominator, $f(x)$ is an improper algebraic fraction and polynomial division must be carried out first. Expanding the denominator:
$(x-2)(x^2+1) = x^3 - 2x^2 + x - 2$. Dividing $2x^4 - 3x^3 + kx^2 - 8x + 3$ by $x^3 - 2x^2 + x - 2$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 15.01 Hypothesis Testing

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1: \mathbf{r} = (1;0;-1) + \lambda(1;2;2), \quad L_2: \mathbf{r} = (2;-4;7) + \mu(2;1;-2).$ The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$ ✓

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

1. **Determine the positions of P and Q:** Let P on L_1 have coordinates $(1+\lambda, 2\lambda, -1+2\lambda)$ and Q on L_2 have coordinates $(2+2\mu, -4+\mu, 7-2\mu)$. The displacement vector $\vec{PQ}$ is:

$\vec{PQ} = ((2+2\mu)-(1+\lambda); (-4+\mu)-2\lambda; (7-2\mu)-(-1+2\lambda)) = (1+2\mu-\lambda; -4+\mu-2\lambda; 8-2\mu-2\lambda).$ Since $PQ \perp L_1$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Exponentials & Logarithms

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2-X)$ m, where X is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, , 0 \le x \le 2, ; 0, , \text{otherwise}. \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$ ✓

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

Let A denote the area of the rectangle and $L_{\min}$ denote the length of the shorter side. 1. **Determine the condition for $A \ge \frac{3}{4}$:** The area is given by $A = X(2-X) = 2X - X^2$.
$2X - X^2 \ge \frac{3}{4} \Leftrightarrow X^2 - 2X + \frac{3}{4} \le 0 \Leftrightarrow (X - \frac{1}{2})(X - \frac{3}{2}) \le 0$ Thus, $A \ge \frac{3}{4}$ corresponds to the interval $E_1 = [\frac{1}{2}, \frac{3}{2}]$. Using the cumulative distribution function.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Trigonometry

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $e^{-2} + 16e^{-4} \approx 0.428$ ✓

b. $16e^{-4} \approx 0.293$

c. $4e^{-2} \approx 0.541$

d. $1 - e^{-1} - \frac{7}{2}e^{-2} \approx 0.158$

A Type II error occurs when H_0 is accepted despite the true parameter value being $\lambda = 2$ (i.e. H_1 is true). Under $\lambda = 2$, the independent counts $X_1, X_2 \sim \text{Po}(2)$. The probability mass function is: $P(X = k) = \frac{2^k e^{-2}}{k!}$ Specifically: $P(X = 0) = e^{-2}, \quad P(X = 1) = 2e^{-2}, \quad P(X = 2) = 2e^{-2}$ H_0 can be accepted in two mutually exclusive ways: 1. **In Stage 1:** H_0 is accepted immediately if $X_1 = 0$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Differentiation

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$ ✓

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on C_2, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at A is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Integration

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. **0.980** ✓

b. 0.900

c. 0.810

d. 0.990

1. **Margin of error for each confidence interval:** For a symmetric 90% confidence interval with known σ, the critical value is $z = 1.645$. Each interval has half-width: $E = 1.645\frac{\sigma}{\sqrt{n}}$ The two intervals are $I_1 = [\bar{X}_1 - E, \bar{X}_1 + E]$ and $I_2 = [\bar{X}_2 - E, \bar{X}_2 + E]$. 2. **Condition for interval overlap:** Two intervals of identical half-width E overlap if and only if the distance between their centres does not exceed the sum of their half-widths.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Sequences & Series

Determine the complete set of real values of the constant k for which the equation $2\cos 2\theta - 4\sin\theta = k$ has at least one real solution for θ.

a. $-6 \le k \le 3$ ✓

b. $-6 \le k \le 2$

c. $k \le 3$

d. $-3 \le k \le 6$

To find the set of values of k for which real solutions exist, we express the left-hand side entirely in terms of $\sin\theta$. Using the double-angle identity $\cos 2\theta = 1 - 2\sin^2\theta$: $2(1 - 2\sin^2\theta) - 4\sin\theta = k$ $-4\sin^2\theta - 4\sin\theta + 2 = k$ Let $u = \sin\theta$. Since $\theta \in \mathbb{R}$, u is constrained to the closed interval $[-1, 1]$. The problem reduces to finding the range of the quadratic function $f(u) = -4u^2 - 4u + 2, \quad u \in [-1, 1]$. 1. **Find the vertex (s.

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